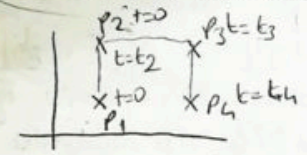
...dönmüyor. # Keyfi Noktaya Gore.

SORU ⇒ Relaxed uc kosullarini kullanarak $P_1[1\ 2]$, $P_2[3\ 3]$, $P_3[2,5]$, $P_4[2\ 1]$ noktalarindan gecen 3 adet kübik spline cizilmek istenmektedir. Kose noktalarinin 1. dereceden türevlerini bulunuz.



$t_2 = |P_1P_2| \Rightarrow \sqrt{0^2+1^2} = 1$

$t_3 = |P_2P_3| \Rightarrow \sqrt{1^2+0^2} = 1$

$t_4 = |P_3P_4| \Rightarrow \sqrt{0+(-2)^2} = 2$

① P_1 için $\Rightarrow P''(0) = 0 \Rightarrow 2B_3 = 0 \rightarrow$ Başlangıç

$P_1' + \frac{1}{2}P_2' = \frac{3}{2}\frac{(P_2-P_1)}{t_2}$

$P(t) = B_1 + B_2 t + B_3 t^2 + B_4 t^3$

② $P''(t_2) = Q''(0)$

$P''(t_2) = 2B_3 + 6B_4 t_2 = 2B_3$

$t_3 \cdot P_1' + 2(t_2+t_3)P_2' + t_2 \cdot P_3' = \frac{3}{t_2 t_3}\left(t_2^2(P_3-P_2) + t_3^2(P_2-P_1)\right)$

③ $P''(t_3) = Q''(0)$

$t_4 \cdot P_2' + 2(t_3+t_4)\cdot P_3' + t_3 \cdot P_4' = \frac{3}{t_3 t_4}\left(t_3^2(P_4-P_3) + t_4^2(P_3-P_2)\right)$

④ P_4 için $\Rightarrow P''(0) = 0 \Rightarrow 2B_3 = 0 \rightarrow$ Bitiş

$2P_3' + 4P_4' = \frac{6}{t_4}(P_4-P_3)$

P_1P_2 $[0\ 1]$ P_4P_3 $[0\ -2]$ P_2P_3 $[1\ 0]$

$$\begin{bmatrix} 1 & 0{,}5 & 0 & \\ t_3 & 2(t_2+t_3) & t_2 & \\ & t_4 & 2(t_3+t_4) & t_3 \\ & & 2 & 4 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \\ P_3' \\ P_4' \end{bmatrix} = \begin{bmatrix} 3/2\ \frac{(P_2-P_1)}{t_2} \\ 3/t_2t_3\left(t_2^2(P_3-P_2)+t_3^2(P_2-P_1)\right) \\ 3/t_3t_4\left(t_3^2(P_4-P_3)+t_4^2(P_3-P_2)\right) \\ 6/t_4\,(P_4-P_3) \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0{,}5 & 0 & \\ 1 & 4 & 1 & \\ & 2 & 6 & 1 \\ & & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} P_1' \\ P_2' \\ P_3' \\ P_4' \end{bmatrix} = \begin{bmatrix} [0\ 1{,}5] \\ [3\ \ 3\ ] \\ [6\ -3] \\ [0\ -6] \end{bmatrix}$$

$P_1' + 0{,}5P_2' = [0\ 1{,}5]$

$P_1' + 4P_2' + P_3' = [3\ \ 3]$

$2P_2' + 6P_3' + P_4' = [6\ -3]$ → 2'ye bölersek

$2P_3' + 4P_4' = [0\ -6]$

$1{,}5P_2' + P_3' = [3\ \ 1{,}5] \Rightarrow$ 3,5'e bölersek

$P_2' + 2P_3' + 0{,}5P_4' = [3\ -1{,}5]$

$P_2' + 0{,}285P_3' = [0{,}857\ 0{,}428]$

$P_2' + 3P_3' + P_4' = [3\ \ -1{,}5]$

$2{,}715P_3' + P_4' = [2{,}143\ -1{,}928]$ ← 2'ye bölersek

$P_3' + 0{,}368P_4' = [0{,}788\ -0{,}710]$

$P_3' + 2P_4' = [0\ \ -5]$

$1{,}632P_4' = [-0{,}788\ -2{,}25]$

$P_4' = [-0{,}483\ -1{,}401]$

$P_3' + 2[-0{,}483\ -1{,}401] = [0\ \ -5]$

$P_3' = [0{,}966\ -0{,}196]$

$2x + y = 5$

$x + y = 6$

$P_2' + 3P_3' + P_4' = [3 \quad -1,5]$

$P_2' + 3[0,966 \quad -0,194] + [-0,683 \quad -1,403] = [3 \quad -1,5]$

$$\boxed{P_2' = [0,585 \quad 0,485]}$$

$P_1' + 0,5 P_2' = [0 \quad 1,5]$

$P_1' + 0,5[0,585 \quad 0,685] = [0 \quad 1,5]$

1,88f

$$\boxed{P_1' = [-0,292 \quad 1,257]}$$

$P_1' = [-1,065 \quad 0,6848]$

$P_2' = [2,131 \quad 1,6304]$

$P_3' = [0,868 \quad -0,1304]$

$P_4' = [-0,4345 \quad -1,4348]$